# Week 18 Self-Assessment

## MCQ Practice - Mixed Review

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# 1 Instructions

This quiz provides comprehensive practice across all topics covered this term. Work through each investigation carefully, as this mirrors the exam format. Test your ability to identify designs, select appropriate tests, and interpret results.

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# 2 Part 1: Conceptual Review

## 2.1 Question 1: Design Types

Match each scenario to its design type:

Different participants in each condition:

- (A) Between-subjects
- (B) Within-subjects
- (C) Correlational
- (D) Case study

Same participants tested in all conditions:

- (A) Between-subjects
- (B) Within-subjects
- (C) Correlational
- (D) Matched pairs

Measuring the relationship between two variables without manipulation:

- (A) Between-subjects
- (B) Within-subjects
- (C) Correlational
- (D) Experimental

## 2.2 Question 2: Test Selection

Two independent groups:

- (A) Independent t-test
- (B) Paired t-test

- (C) Pearson's r
- (D) Chi-square

Same participants, two time points:

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson's r
- (D) ANOVA

Relationship between two continuous variables:

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson's r
- (D) Chi-square

### 2.3 Question 3: Statistical Significance

$p < .05$ means the result is:

- (A) Definitely true
- (B) Statistically significant at alpha = .05
- (C) Practically important
- (D) 100% reliable

$p = .08$ means:

- (A) Significant at .05
- (B) Not significant at .05
- (C) Definitely no effect
- (D) Large effect

### 2.4 Question 4: Effect Sizes

Cohen's $d = 0.2$ is:

- (A) Small
- (B) Medium
- (C) Large
- (D) Negligible

Cohen's $d = 0.5$ is:

- (A) Small
- (B) Medium
- (C) Large
- (D) Very large

Cohen's $d = 0.8$ is:

- (A) Small

- (B) Medium
- (C) Large
- (D) Massive

### 2.5 Question 5: Correlation Interpretation

$r = .65$ indicates:

- (A) Weak positive
- (B) Moderate-strong positive
- (C) Weak negative
- (D) Strong negative

$r = -.40$ indicates:

- (A) Weak positive
- (B) Strong positive
- (C) Moderate negative
- (D) No relationship

### 2.6 Question 6: Key Concepts

Correlation implies causation: TRUE / FALSE

Effect size tells us practical importance: TRUE / FALSE

A 95% CI that excludes zero suggests significance at $p < .05$: TRUE / FALSE

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## 3 Part 2: Full Investigation Practice

### 3.1 Investigation 1

A researcher tests whether background music affects concentration. 60 participants are randomly assigned to work in silence or with classical music playing. Concentration is measured using an error-counting task.

1. **Design type:**

- (A) Between-subjects
- (B) Within-subjects
- (C) Correlational
- (D) Matched pairs

2. **IV:**

- (A) Background condition (silence vs music)
- (B) Concentration score
- (C) Number of errors
- (D) No IV

3. **DV:**

- (A) Music condition
- (B) Concentration (error count)
- (C) Participant preference
- (D) Room assignment

4. **Appropriate test:**

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson's r
- (D) Chi-square

5. **One-tailed hypothesis:**

- (A) Music will affect concentration
- (B) Participants working in silence will make fewer errors than those with music
- (C) There is no effect of music
- (D) Errors and music are correlated

6. **Results: $t(58) = 2.34$, $p = .023$, $d = 0.61$. Significant at alpha = .05?** TRUE / FALSE

7. **Effect size:**

- (A) Small
- (B) Medium
- (C) Large
- (D) Not meaningful

## 3.2 Investigation 2

A researcher measures stress levels before and after a relaxation intervention in 25 participants.

1. **Design type:**

- (A) Between-subjects
- (B) Within-subjects
- (C) Correlational
- (D) Case study

2. **IV:**

- (A) Time (before vs after relaxation)
- (B) Stress level
- (C) Relaxation technique
- (D) No IV

3. **DV:**

- (A) Time point
- (B) Stress level

- (C) Intervention type
- (D) Participant anxiety

4. **Appropriate test:**

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson's r
- (D) ANOVA

5. **Results: $t(24) = -3.89$, $p < .001$, $d = 0.78$. The negative t-value indicates:**

- (A) Error in analysis
- (B) Stress decreased (before was higher than after)
- (C) Not significant
- (D) Positive effect

6. **Effect size:**

- (A) Small
- (B) Medium-large
- (C) Very small
- (D) Cannot determine

7. **Conclusion:**

- (A) Stress significantly decreased after relaxation, medium-large effect
- (B) No change in stress
- (C) Stress increased
- (D) Inconclusive

### 3.3 Investigation 3

A researcher examines whether screen time is related to sleep quality in 80 teenagers.

1. **Design type:**

- (A) Between-subjects
- (B) Within-subjects
- (C) Correlational
- (D) Quasi-experimental

2. **Is there an IV?**

- (A) Yes, screen time
- (B) Yes, sleep quality
- (C) No IV - both variables are measured
- (D) Yes, age

3. **Appropriate test:**

- (A) Independent t-test

- (B) Paired t-test
- (C) Pearson's r
- (D) Chi-square

4. **Results: $r(78) = -.42$, $p < .001$. Direction:**

- (A) Positive
- (B) Negative (more screen time, worse sleep)
- (C) No relationship
- (D) Cannot tell

5. **Strength:**

- (A) Weak
- (B) Moderate
- (C) Strong
- (D) Perfect

6. **Can we conclude screen time CAUSES poor sleep?** TRUE / FALSE

7. **Name a possible third variable:**

- (A) Stress, depression, or poor time management
- (B) The correlation itself
- (C) Sample size
- (D) p-value

## 3.4 Investigation 4

A researcher compares reaction times between morning people (n=35) and evening people (n=35), tested at 9am.

1. **Design type:**

- (A) Between-subjects
- (B) Within-subjects
- (C) Correlational
- (D) Repeated measures

2. **IV:**

- (A) Chronotype (morning vs evening person)
- (B) Reaction time
- (C) Test time
- (D) No IV

3. **DV:**

- (A) Chronotype
- (B) Reaction time
- (C) Time of testing

- (D) Participant preference

4. **Appropriate test:**

- (A) Independent t-test
- (B) Paired t-test
- (C) Correlation
- (D) Regression

5. **Results: $t(68) = 3.12$, $p = .003$, $d = 0.75$. Effect size:**

- (A) Small
- (B) Medium-large
- (C) Negligible
- (D) Cannot interpret

6. **Morning people had faster reaction times. Is this causal?**

- (A) We can suggest it, but this is quasi-experimental (chronotype not randomly assigned)
- (B) Yes, definitively causal
- (C) No relationship exists
- (D) Cannot interpret without more data

## 3.5 Investigation 5

A regression analysis predicts exam performance from hours of revision: Performance = 45 + 3.5(Hours), R2 = .38

1. **This analysis allows us to:**

- (A) Compare groups
- (B) Predict performance from revision hours
- (C) Determine causation
- (D) Test for differences

2. **The intercept (45) means:**

- (A) Predicted performance when revision hours = 0
- (B) The slope
- (C) R-squared
- (D) Maximum performance

3. **Each additional hour of revision is associated with:**

- (A) 45 more points
- (B) 3.5 more points
- (C) 38% increase
- (D) No change

4. **If someone revises for 10 hours, predicted performance is:**

- (A) 45

- (B) 35
- (C) 100
- (D) 80

5. **R2 = .38 means:**

- (A) 38% of variance in performance is explained by revision
- (B) The correlation is .38
- (C) 38% of students improved
- (D) Prediction is 38% accurate

6. **The remaining 62% of variance is:**

- (A) Due to other factors (ability, sleep, test anxiety, etc.)
- (B) Measurement error only
- (C) Proof the model is wrong
- (D) Impossible to explain

## 3.6 Investigation 6

A study reports a 95% CI for the mean difference between two groups: [2.5, 8.3].

1. **Does this CI include zero?** TRUE / FALSE

2. **This suggests the difference is:**

- (A) Not significant
- (B) Statistically significant at $p < .05$
- (C) Very large
- (D) Zero

3. **The point estimate for the difference is approximately:**

- (A) 2.5
- (B) 8.3
- (C) 5.4 (midpoint of the interval)
- (D) 0

4. **A narrower CI would indicate:**

- (A) Larger effect
- (B) Greater precision in the estimate
- (C) Non-significance
- (D) Smaller sample

5. **If another study found CI [−1.2, 4.8] for the same comparison:**

- (A) That result would not be significant (CI includes zero)
- (B) Both studies agree
- (C) The effect is larger
- (D) We should average them

# 4 Part 3: Quick-Fire Review

## 4.1 Research Credibility

Pre-registration helps prevent:

- (A) Participant dropout
- (B) p-hacking and HARKing
- (C) Data collection errors
- (D) Small samples

HARKing means:

- (A) Honest research
- (B) Hypothesising After Results are Known
- (C) High-quality analysis
- (D) Helpful research knowledge

Publication bias refers to:

- (A) Journals being unfair
- (B) Significant results being more likely to be published
- (C) Poor peer review
- (D) Too many publications

## 4.2 Effect Size vs Significance

A result can be significant but have a small effect size: TRUE / FALSE

A large effect size guarantees statistical significance: TRUE / FALSE

Effect size is independent of sample size: TRUE / FALSE

p-value depends partly on sample size: TRUE / FALSE

## 4.3 Degrees of Freedom

Independent t-test, n1=20, n2=20: df =

- (A) 40
- (B) 38
- (C) 20
- (D) 19

Paired t-test, n=30: df =

- (A) 30
- (B) 29
- (C) 28
- (D) 60

Correlation, n=50: df =

- (A) 50
- (B) 49
- (C) 48
- (D) 100

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## 5 Summary Checklist

Before the exam, verify you can:

- ☐ Identify design type from any research description
- ☐ Identify IV and DV (or recognise no IV in correlational designs)
- ☐ Select the appropriate statistical test
- ☐ Write one-tailed and two-tailed hypotheses
- ☐ Interpret p-values correctly
- ☐ Interpret effect sizes (small/medium/large)
- ☐ Interpret confidence intervals
- ☐ Interpret regression output (slope, intercept, R2)
- ☐ Explain why correlation does not imply causation
- ☐ Identify questionable research practices

> ! Exam Preparation
>
> Review your previous quiz attempts. Focus on areas where you made errors. Practice identifying designs and tests quickly - this is the foundation for all other questions.